

EXERCISE 6.7

Q.1 Find equations of the tangent and normal to each of the following at the indicated point

(i) $y^2 = 4ax$ at $(at^2, 2at)$

Solution:

$$y^2 = 4ax \dots (i)$$

Replace $x = \frac{x+x_1}{2}, y^2 = yy_1$ in (i)

$$yy_1 = 4a \left(\frac{x+x_1}{2} \right)$$

$$yy_1 = 2a(x+x_1)$$

Put $x_1 = at^2, y_1 = 2at$, we get;

$$y(2at) = 2a(x+at^2)$$

$$yt = x + at^2$$

$$\boxed{y = \frac{1}{t}x + at} \text{ is equation of tangent}$$

Slope of tangent line is

$$m_1 = \frac{1}{t}$$

Slope of normal is

$$m = -\frac{1}{m_1}$$

$$m = -t$$

Equation of normal passing through

$P(at^2, 2at)$ with slope $m = -t$ is

given by

$$y - y_1 = m(x - x_1)$$

$$y - 2at = -t(x - at^2)$$

$$\boxed{y = -tx + at^3 + 2at}$$

(ii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $(a \cos \theta, b \sin \theta)$

Solution:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \dots (i)$$

Replace x^2 by xx_1 and y^2 by yy_1 in

(i)

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$$

Put $x_1 = a \cos \theta, y_1 = b \sin \theta$, we get;

$$\frac{(a \cos \theta)x}{a^2} + \frac{(b \sin \theta)y}{b^2} = 1$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

$$\boxed{x(b \cos \theta) + y(a \sin \theta) - ab = 0}$$

is equation of tangent

$$\text{Slope of tangent is } m_1 = \frac{-b \cos \theta}{a \sin \theta}$$

$$m_1 = -\frac{b}{a} \cot \theta$$

Slope of normal is

$$m = -\frac{1}{m_1} = -\frac{1}{-\frac{b}{a} \cot \theta} = \frac{a}{b} \tan \theta$$

Equation of normal passing

through $(a \cos \theta, b \sin \theta)$ with slope

$$m = \frac{a}{b} \tan \theta \text{ is}$$

given by

$$y - y_1 = m(x - x_1)$$

$$y - b \sin \theta = \frac{a}{b} \tan \theta (x - a \cos \theta)$$

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

$$b \cos \theta (y - b \sin \theta) = a \sin \theta (x - a \cos \theta)$$

$$(b \cos \theta)y - b^2 \cos \theta \sin \theta = (a \sin \theta)x - a^2 \cos \theta \sin \theta$$

$$(a \sin \theta)x - (b \cos \theta)y - a^2 \cos \theta \sin \theta + b^2 \cos \theta \sin \theta = 0$$

Dividing by $\cos \theta \sin \theta$ we get;

$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} - a^2 + b^2 = 0$$

$$\boxed{ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2}$$

(iii) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $(a \sec \theta, b \tan \theta)$

Solution:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \dots (i)$$

Replace x^2 by xx_1 and y^2 by yy_1

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

Put $x_1 = a \sec \theta, y_1 = b \tan \theta$

$$\frac{x(a \sec \theta)}{a^2} - \frac{y(b \tan \theta)}{b^2} = 1$$

$$\boxed{\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1}$$

is equation of tangent

Slope of tangent line is

$$m_1 = \frac{-\frac{\sec \theta}{a}}{\frac{\tan \theta}{b}}$$

$$m_1 = \frac{\sec \theta}{a} \times \frac{b}{\tan \theta} = \frac{b}{a} \sec \theta \times \cot \theta$$

$$m_1 = \frac{b}{a} \frac{1}{\cos \theta} \cdot \frac{\cos \theta}{\sin \theta} = \frac{b}{a} \cdot \operatorname{cosec} \theta$$

Slope of normal is

$$m = -\frac{1}{m_1} = \frac{-a}{b \operatorname{cosec} \theta} = \frac{-a \sin \theta}{b}$$

Equation of the normal passing through $(a \sec \theta, b \tan \theta)$ with

$$m = -\frac{a}{b} \sin \theta \text{ is given by}$$

$$y - y_1 = m(x - x_1)$$

$$y - b \tan \theta = \frac{-a \sin \theta}{b} (x - a \sec \theta)$$

$$by - b^2 \tan \theta = -a \sin \theta x + a^2 \sin \theta \sec \theta$$

$$(a \sin \theta)x + by = \frac{a^2 \sin \theta}{\cos \theta} + b^2 \tan \theta$$

$$(a \sin \theta)x + by = (a^2 + b^2) \tan \theta$$

Dividing by $\tan \theta$ on both sides we get;

$$\left(\frac{a \sin \theta}{\tan \theta} \right) x + \frac{by}{\tan \theta} = a^2 + b^2$$

$$\boxed{(a \cos \theta)x + (b \cot \theta)y = a^2 + b^2}$$

Q.2 Write equation of the tangent to the given conic at the indicated point

(i) $3x^2 = -16y$ at the points whose ordinate is -3

Solution:

$$3x^2 = -16y \dots (i)$$

Put ordinate, $y = -3$ in (i)

$$3x^2 = -16(-3)$$

$$3x^2 = 48$$

$$x^2 = 16$$

$$x = \pm 4$$

Hence points of contact are $A(4, -3)$

and $B(-4, -3)$

Now for equation of tangent, we

replace x^2 by xx_1 , y by $\frac{y+y_1}{2}$

$$3xx_1 = -\left(\frac{y+y_1}{2}\right)$$

$$3xx_1 = -8(y+y_1)$$

Hence equation of tangent at

$A(4, -3)$ is

$$3x(4) = -8(y-3)$$

$$3x = -2(y-3)$$

$$\boxed{3x + 2y - 6 = 0}$$

Equation of tangent at $B(-4, -3)$ is

$$3x(-4) = -8(y-3)$$

$$3x = 2(y-3)$$

$$\boxed{3x - 2y + 6 = 0}$$

(ii) $3x^2 - 7y^2 = 20$ at the points where $y = -1$

Solution:

$$3x^2 - 7y^2 = 20 \dots (i)$$

Put $y = -1$ in (1), we get;

$$3x^2 - 7(-1)^2 = 20$$

$$3x^2 - 7 = 20$$

$$3x^2 = 27$$

$$x^2 = 9$$

$$\Rightarrow x = \pm 3$$

Hence points of contact are

$$A(3, -1), B(-3, -1)$$

For equation of tangent, put

$$x^2 = xx_1, y^2 = yy_1 \text{ in (i)}$$

$$3xx_1 - 7yy_1 = 20$$

So equation of tangent at $A(3, -1)$ is

$$3x(3) - 7y(-1) = 20$$

$$\boxed{9x + 7y = 20}$$

Equation of tangent at $B(-3, -1)$ is

$$3(-3)x - 7y(-1) = 20$$

$$-9x + 7y = 20$$

$$\boxed{9x - 7y + 20 = 0}$$

- (iii) $3x^2 - 7y^2 + 2x - y - 48 = 0$ at the points where $x = 4$

Solution:

$$3x^2 - 7y^2 + 2x - y - 48 = 0 \dots (i)$$

Put $x = 4$ in (i)

$$3(4)^2 - 7y^2 + 2(4) - y - 48 = 0$$

$$48 - 7y^2 + 8 - y - 48 = 0$$

$$-7y^2 - y + 8 = 0$$

$$7y^2 + y - 8 = 0$$

$$7y^2 + 8y - 7y - 8 = 0$$

$$y(7y + 8) - 1(7y + 8) = 0$$

$$(7y + 8)(y - 1) = 0$$

$$7y + 8 = 0, y - 1 = 0$$

$$y = \frac{-8}{7}, y = 1$$

Hence points of contact are

$$A\left(4, \frac{-8}{7}\right), B(4, 1)$$

For equation of tangent, replace

$$x^2 \text{ by } xx_1, y^2 \text{ by } yy_1, x \text{ by } \frac{x+x_1}{2}$$

$$\text{and } y \text{ by } \frac{y+y_1}{2} \text{ in given equation}$$

$$3xx_1 - 7yy_1 + 2\left(\frac{x+x_1}{2}\right) - \left(\frac{y+y_1}{2}\right) - 48 = 0$$

$$3xx_1 - 7yy_1 + (x+x_1) - \left(\frac{y+y_1}{2}\right) - 48 = 0$$

At $A\left(4, \frac{-8}{7}\right)$ equation of tangent is

$$3x(4) - 7y\left(\frac{-8}{7}\right) + (x+4) - \frac{1}{2}\left(y - \frac{8}{7}\right) - 48 = 0$$

$$12x + 8y + x + 4 - \frac{y}{2} + \frac{4}{7} - 48 = 0$$

$$13x + \frac{15}{2}y - \frac{304}{7} = 0$$

$$\boxed{182x + 105y - 608 = 0}$$

At $B(4, 1)$ equation of tangent is

$$3x(4) - 7y(1) + (x+4) - \frac{1}{2}(y+1) - 48 = 0$$

$$12x - 7y + x + 4 - \frac{1}{2}y - \frac{1}{2} - 48 = 0$$

$$24x - 14y + 2x + 8 - y - 1 - 96 = 0$$

$$\boxed{26x - 15y - 89 = 0}$$

Q.3 Find equations of the tangent to each of the following through the given point:

- (i) $x^2 + y^2 = 25$ through $(7, -1)$

Solution:

We can check that point $(7, -1)$ does

not satisfy the given equation

so equation of the tangent is

$$y = mx \pm a\sqrt{1+m^2}$$

Put $x = 7, y = -1$ and $a = 5$

(radius of circle), we get;

$$-1 = 7m \pm 5\sqrt{1+m^2}$$

$$-1 - 7m = \pm 5\sqrt{1+m^2} \dots (i)$$

Squaring on both sides, we get;

$$(-1 - 7m)^2 = (\pm 5\sqrt{1+m^2})^2$$

$$1 + 49m^2 + 14m = 25(1+m^2)$$

$$1 + 49m^2 + 14m = 25 + 25m^2$$

$$24m^2 + 14m - 24 = 0$$

$$12m^2 + 7m - 12 = 0$$

$$m = \frac{-7 \pm \sqrt{(7)^2 - 4(12)(-12)}}{2(12)}$$

$$m = \frac{-7 \pm \sqrt{49 + 576}}{24}$$

$$m = \frac{-7 \pm \sqrt{625}}{24}$$

$$m = \frac{-7 \pm 25}{24}$$

$$m = \frac{-7 + 25}{24}, \quad m = \frac{-7 - 25}{24}$$

$$= \frac{18}{24}, \quad = \frac{-32}{24}$$

$$m = \frac{3}{4}, \quad m = \frac{-4}{3}$$

$$m = \frac{3}{4}, -\frac{4}{3}$$

When $m = \frac{3}{4}$, equation (i)

$-1 - 7m = -5\sqrt{1 + m^2}$ is satisfied,
equation of tangent is

$$y = mx - a\sqrt{1 + m^2}$$

$$y = \frac{3}{4}x - 5\sqrt{1 + \frac{9}{16}}$$

$$y = \frac{3}{4}x - 5\sqrt{\frac{25}{16}}$$

$$y = \frac{3}{4}x - 5 \times \frac{5}{4}$$

$$y = \frac{3}{4}x - \frac{25}{4}$$

$$4y = 3x - 25$$

$$\boxed{3x - 4y - 25 = 0}$$

When $m = \frac{-4}{3}$, equation (i)

$-1 - 7m = 5\sqrt{1 + m^2}$ is satisfied,
equation of tangent is

$$y = mx + a\sqrt{1 + m^2}$$

$$y = \frac{-4}{3}x + 5\sqrt{1 + \frac{16}{9}}$$

$$y = \frac{-4}{3}x + 5\sqrt{\frac{25}{9}}$$

$$y = \frac{-4}{3}x + 5 \times \frac{5}{3}$$

$$3y = -4x + 25$$

$$\boxed{4x + 3y - 25 = 0}$$

(ii) $y^2 = 12x$ through (1,4)

Solution:

$$y^2 = 12x$$

Here $4a = 12 \Rightarrow a = 3$

Equation of tangent is

$$y = mx + \frac{a}{m} \dots (i)$$

Put $x = 1, y = 4, a = 3$ in (i)

$$4 = m + \frac{3}{m}$$

$$4 = \frac{m^2 + 3}{m}$$

$$4m = m^2 + 3$$

$$m^2 - 4m + 3 = 0$$

$$m^2 - 3m - m + 3 = 0$$

$$m(m - 3) - 1(m - 3) = 0$$

$$(m - 1)(m - 3) = 0$$

$$m - 1 = 0, m - 3 = 0$$

$$m = 1, m = 3$$

When $m = 1$ then (i) becomes

$$\boxed{y = x + 3}$$

When $m = 3$ then (i) becomes

$$y = 3x + \frac{3}{3}$$

$$\boxed{y = 3x + 1}$$

(iii) $x^2 - 2y^2 = 2$ through (1,-2)

Solution:

$$x^2 - 2y^2 = 2$$

$$\frac{x^2}{2} - y^2 = 1$$

$$\Rightarrow a^2 = 2, b^2 = 1$$

Equations of tangents for hyperbola are given by

$$y = mx \pm \sqrt{a^2 m^2 - b^2} \dots (i)$$

Put $x = 1, y = -2, a^2 = 2, b^2 = 1$, in (i) we get;

$$-2 = m \pm \sqrt{2m^2 - 1}$$

$$-2 - m = \pm \sqrt{2m^2 - 1}$$

$$-(2 + m) = \pm \sqrt{2m^2 - 1} \dots (ii)$$

Squaring on both sides we get;

$$4 + m^2 + 4m = 2m^2 - 1$$

$$2m^2 - m^2 - 4m - 4 - 1 = 0$$

$$m^2 - 4m - 5 = 0$$

$$m^2 - 5m + m - 5 = 0$$

$$m(m - 5) + 1(m - 5) = 0$$

$$(m + 1)(m - 5) = 0$$

$$m + 1 = 0, m - 5 = 0$$

$$m = -1, 5$$

When $m = -1$, equation (ii)

$$-(2 + m) = -\sqrt{2m^2 - 1} \text{ is satisfied}$$

Equation of tangent is

$$y = mx - \sqrt{2m^2 - 1}$$

$$y = -x - \sqrt{2(-1)^2 - 1}$$

$$y = -x - \sqrt{2 - 1}$$

$$y = -x - 1$$

$$\boxed{x + y + 1 = 0}$$

When $m = 5$, equation (ii)

$$-(2 + m) = -\sqrt{2m^2 - 1} \text{ is satisfied}$$

Equation of tangent is

$$y = mx - \sqrt{2m^2 - 1}$$

$$y = 5x - \sqrt{2(5)^2 - 1}$$

$$y = 5x - 7$$

$$\boxed{5x - y - 7 = 0}$$

Q.4 Find equations of the Normals to the parabola $y^2 = 8x$ which are parallel to the line $2x + 3y = 10$

Solution:

given equation of line is

$$2x + 3y - 10 = 0 \dots (i)$$

Slope of the given line is $m_1 = -\frac{2}{3}$

As required normal lines are parallel to (i), so slope of normal line is

$$m = m_1 = -\frac{2}{3}$$

$$y^2 = 8x \dots (ii)$$

Differentiate w.r.t. x

$$2yy' = 8$$

$$y' = \frac{4}{y}$$

At (x_1, y_1) , $y' = \frac{4}{y_1} = \text{slope of tangent}$

Slope of normal at $(x_1, y_1) = -\frac{y_1}{4}$

But given slope of normal is $-\frac{2}{3}$

$$\text{Hence } -\frac{y_1}{4} = -\frac{2}{3}$$

$$y_1 = \frac{8}{3}$$

Since (x_1, y_1) lies on (ii) so

$$y_1^2 = 8x_1$$

$$\left(\frac{8}{3}\right)^2 = 8x_1 \quad \because y_1 = \frac{8}{3}$$

$$\frac{64}{9} = 8x_1$$

$$x_1 = \frac{8}{9}$$

So point of intersection is $\left(\frac{8}{9}, \frac{8}{3}\right)$

Hence required equation of normal
to (ii) at $\left(\frac{8}{9}, \frac{8}{3}\right)$ with slope $m = -\frac{2}{3}$

is

$$\begin{aligned}y - y_1 &= m(x - x_1) \\y - \frac{8}{3} &= -\frac{2}{3}\left(x - \frac{8}{9}\right) \\y - \frac{8}{3} &= -\frac{2}{3}\left(\frac{9x - 8}{9}\right)\end{aligned}$$

Multiply with 27 we get;

$$27y - 72 = -2(9x - 8)$$

$$27y - 72 = -18x + 16$$

$$\boxed{18x + 27y - 88 = 0}$$

Q.5 Find equations the tangents to the ellipse $\frac{x^2}{4} + y^2 = 1$ which are parallel to the line $2x - 4y + 5 = 0$

Solution:

$$\frac{x^2}{4} + y^2 = 1 \dots (i)$$

$$\Rightarrow a^2 = 4, b^2 = 1$$

Equation of tangents to (i) are given by

$$y = mx \pm \sqrt{a^2 m^2 + b^2}$$

Put $a^2 = 4, b^2 = 1$ we get;

$$y = mx \pm \sqrt{4m^2 + 1} \dots (ii)$$

As (ii) are parallel to $2x - 4y + 5 = 0$,

so slope of tangent line is $m = \frac{1}{2}$

Put $m = \frac{1}{2}$ in (ii), we get required

equations of tangent lines

$$y = \frac{1}{2}x \pm \sqrt{4\left(\frac{1}{2}\right)^2 + 1}$$

$$y = \frac{1}{2}x \pm \sqrt{4\left(\frac{1}{4}\right) + 1}$$

$$y = \frac{1}{2}x \pm \sqrt{2}$$

$$2y = x \pm 2\sqrt{2}$$

$$x - 2y \pm 2\sqrt{2} = 0$$

Hence required equations of tangents are

$$\boxed{x - 2y + 2\sqrt{2} = 0}$$

$$\boxed{x - 2y - 2\sqrt{2} = 0}$$

Q.6 Find equations of the tangents to the conic $9x^2 - 4y^2 = 36$ parallel to $5x - 2y + 7 = 0$

Solution:

$$9x^2 - 4y^2 = 36$$

Dividing by 36 on both sides, we get;

$$\frac{x^2}{4} - \frac{y^2}{9} = 1 \dots (i)$$

$$\Rightarrow a^2 = 4, b^2 = 9$$

Equations of tangents to (i) are

$$y = mx \pm \sqrt{a^2 m^2 - b^2}$$

$$y = mx \pm \sqrt{4m^2 - 9} \dots (ii)$$

As (ii) are parallel to $5x - 2y + 7 = 0$

So slope of (ii) is $m = \frac{5}{2}$

Hence equations of tangents are given by

$$y = \frac{5}{2}x \pm \sqrt{4\left(\frac{5}{2}\right)^2 - 9}$$

$$y = \frac{5}{2}x \pm \sqrt{4\left(\frac{25}{4}\right) - 9}$$

$$y = \frac{5}{2}x \pm \sqrt{25 - 9}$$

$$y = \frac{5}{2}x \pm \sqrt{16}$$

$$y = \frac{5}{2}x \pm 4$$

$$2y = 5x \pm 8$$

$$5x - 2y \pm 8 = 0$$

Hence required equations of tangents are

$$\boxed{5x - 2y + 8 = 0, 5x - 2y - 8 = 0}$$

Q.7 Find equations of the common tangents to the given conics

(i) $x^2 = 80y$ and $x^2 + y^2 = 81$

Solution:

$$x^2 = 80y \dots (i)$$

Equation of tangent is

$$y = mx - am^2$$

$$y = mx - 20m^2 \quad (4a = 80 \Rightarrow a = 20)$$

$$x^2 + y^2 = 81 \dots (ii)$$

Equation of tangent is

$$y = mx \pm a\sqrt{m^2 + 1}$$

$$y = mx \pm 9\sqrt{m^2 + 1} \quad (a^2 = 81 \Rightarrow a = 9)$$

For a common tangent, we must have

$$9\sqrt{m^2 + 1} = -20m^2$$

Squaring on both sides we get;

$$81(m^2 + 1) = 400m^4$$

$$400m^4 - 81m^2 - 81 = 0$$

$$m^2 = \frac{-(-81) \pm \sqrt{(-81)^2 - 4(400)(-81)}}{2(400)}$$

$$m^2 = \frac{81 \pm \sqrt{6561 + 129600}}{800}$$

$$m^2 = \frac{81 \pm \sqrt{136161}}{800}$$

$$m^2 = \frac{81 \pm 369}{800}$$

$$m^2 = \frac{81 + 369}{800}, m^2 = \frac{81 - 369}{800}$$

$$m^2 = \frac{450}{800}, m^2 = -\frac{288}{800}$$

$$m^2 = \frac{9}{16}, \text{ neglect } m^2 = \frac{-288}{800}$$

because m^2 cannot be negative

$$m = \pm \frac{3}{4}$$

Hence equations of the common tangents to (i) are

$$y = mx - 20m^2$$

$$y = \pm \frac{3}{4}x - 20\left(\frac{9}{16}\right)$$

$$y = \pm \frac{3}{4}x - \frac{45}{4}$$

$$4y = \pm 3x - 45$$

$$\boxed{\pm 3x - 4y - 45 = 0}$$

(ii) $y^2 = 16x$ and $x^2 = 2y$

Solution:

$$y^2 = 16x \dots (i)$$

Equation of tangent is

$$y = mx + \frac{a}{m}$$

$$y = mx + \frac{4}{m} \dots (ii) \quad \because (4a = 16 \Rightarrow a = 4)$$

and

$$x^2 = 2y \dots (iii)$$

Equation of tangent is

$$y = mx - am^2$$

$$y = mx - \frac{1}{2}m^2 \dots (iv) \quad \because (4a = 2 \Rightarrow a = \frac{1}{2})$$

For a common tangent, we must have

$$\frac{4}{m} = -\frac{1}{2}m^2$$

$$\frac{8}{m} = -m^2$$

$$m^2 + \frac{8}{m} = 0$$

$$m^3 + 8 = 0$$

$$(m + 2)(m^2 - 4m + 4) = 0$$

$$m + 2 = 0, m^2 - 4m + 4 = 0$$

$$m = -2 \quad (m^2 - 4m + 4 = 0 \text{ gives imaginary roots})$$

Put $m = -2$ in (ii)

$$y = -2x + \frac{4}{-2}$$

$$y = -2x - 2$$

$$\boxed{2x + y + 2 = 0}$$

Q.8 Find the points of intersection of given conics

(i) $\frac{x^2}{18} + \frac{y^2}{8} = 1$ and $\frac{x^2}{3} - \frac{y^2}{3} = 1$

Solution: Given conics are

$$\frac{x^2}{18} + \frac{y^2}{8} = 1 \dots (i)$$

$$\frac{x^2}{3} - \frac{y^2}{3} = 1 \dots (ii)$$

Multiply (ii) by $\frac{1}{6}$ and then subtracting from (i) we get;

$$\frac{\cancel{x^2}}{18} + \frac{y^2}{8} = 1$$

$$\frac{\pm \cancel{x^2}}{18} \mp \frac{y^2}{18} = \pm \frac{1}{6}$$

$$\frac{13}{72}y^2 = \frac{5}{6}$$

$$y^2 = \frac{60}{13}$$

$$y = \pm \sqrt{\frac{60}{13}}$$

Put $y = \pm \sqrt{\frac{60}{13}}$ in (i) we get;

$$\frac{1}{18}x^2 + \frac{1}{8}\left(\frac{60}{13}\right) = 1$$

$$\frac{1}{18}x^2 + \frac{15}{26} = 1$$

$$\frac{1}{18}x^2 = 1 - \frac{15}{26}$$

$$\frac{1}{18}x^2 = \frac{11}{26}$$

$$x^2 = \frac{99}{13} \Rightarrow x = \pm \sqrt{\frac{99}{13}}$$

Hence points of intersection are

$$\left(\pm \sqrt{\frac{99}{13}}, \pm \sqrt{\frac{60}{13}} \right)$$

(ii) $x^2 + y^2 = 8$ and $x^2 - y^2 = 1$

Solution:

Given conics are

$$x^2 + y^2 = 8 \dots (i)$$

$$x^2 - y^2 = 1 \dots (ii)$$

Adding (i) and (ii) we get;

$$2x^2 = 9 \Rightarrow x^2 = \frac{9}{2}$$

$$x = \pm \frac{3}{\sqrt{2}}$$

Put $x = \pm \frac{3}{\sqrt{2}}$ in (i) we get;

$$\frac{9}{2} + y^2 = 8 \Rightarrow y^2 = 8 - \frac{9}{2}$$

$$y^2 = \frac{7}{2}$$

$$y = \pm \sqrt{\frac{7}{2}}$$

Hence points of intersection are

$$\left(\pm \frac{3}{\sqrt{2}}, \pm \sqrt{\frac{7}{2}} \right)$$

(iii) $3x^2 - 4y^2 = 12$ and $3y^2 - 2x^2 = 7$

Solution:

$$3x^2 - 4y^2 = 12 \dots (i)$$

$$-2x^2 + 3y^2 = 7 \dots (ii)$$

Multiply equation (i) by 2 and (ii) by 3 then by adding the equations, we get;

$$6x^2 - 8y^2 = 24$$

$$\frac{-6x^2 + 9y^2 = 21}{y^2 = 45}$$

$$y = \pm 3\sqrt{5}$$

Put $y = \pm 3\sqrt{5}$ in (i) we get;

$$3x^2 - 4(45) = 12$$

$$3x^2 = 192$$

$$x^2 = 64$$

$$x = \pm 8$$

Hence points of intersection are

$$(\pm 8, \pm 3\sqrt{5})$$

(iv) $3x^2 + 5y^2 = 60$ and $9x^2 + y^2 = 124$

Solution:

$$3x^2 + 5y^2 = 60 \dots (i)$$

$$9x^2 + y^2 = 124 \dots (ii)$$

Multiply (i) by 3 and then
subtracting (ii) from it

$$9x^2 + 15y^2 = 180$$

$$\begin{array}{r} \pm 9x^2 \pm y^2 = \pm 124 \\ \hline 14y^2 = 56 \end{array}$$

$$y^2 = 4$$

$$y = \pm 2$$

Put $y = \pm 2$ in (i) we get;

$$3x^2 + 5(\pm 2)^2 = 60$$

$$3x^2 + 5(4) = 60$$

$$3x^2 = 40$$

$$x^2 = \frac{40}{3}$$

$$x = \pm 2\sqrt{\frac{10}{3}}$$

Hence points of intersections are

$$\left(\pm \sqrt{\frac{40}{3}}, \pm 2 \right)$$

(v) $4x^2 + y^2 = 16$ and

$$x^2 + y^2 + 2y - 8 = 0$$

Solution: Given conics are

$$4x^2 + y^2 = 16 \dots (i)$$

$$x^2 + y^2 + 2y - 8 = 0 \dots (ii)$$

Subtracting (ii) from (i) we get;

$$4x^2 - y^2 = 16$$

$$-x^2 - y^2 + 2y = 8$$

$$\begin{array}{r} 4x^2 - y^2 = 16 \\ \hline -x^2 - y^2 + 2y = 8 \end{array}$$

$$3x^2 = \frac{2y+8}{3} \dots (iii)$$

Put in (i) we get;

$$4\left(\frac{2y+8}{3}\right) + y^2 = 16$$

$$8y + 32 + 3y^2 = 48$$

$$3y^2 + 8y - 16 = 0$$

$$\Rightarrow y = \frac{-8 \pm \sqrt{(8)^2 - 4(3)(-16)}}{2(3)}$$

$$y = \frac{8 \pm \sqrt{256}}{6}$$

$$\Rightarrow \boxed{y = -4, \frac{4}{3}}$$

Put $y = -4$ in (iii)

$$x^2 = 0 \Rightarrow x = 0$$

Put $y = \frac{4}{3}$ in (iii)

$$x^2 = \frac{32}{9}$$

$$\Rightarrow x = \pm \frac{4\sqrt{2}}{3}$$

Hence the points of intersection are

$$(0, -4) \text{ and } \left(\frac{\pm 4\sqrt{2}}{3}, \frac{4}{3} \right)$$